

Decision: **rate manuscript once again after major changes and another review**

July 12, 2025

JURI-00032-2025-01

XGBoost-Based Prediction of pKa Values from Molecular Descriptors

Dear Dr. Al-Khater,

We have carefully evaluated your manuscript, entitled: XGBoost-Based Prediction of pKa Values from Molecular Descriptors. We may be able to accept it if you can respond adequately to the points raised during the review process (see below).

Please revise your manuscript strictly according to the attached Reviewers' comments. Your manuscript won't be taken into consideration without the revisions made according to the recommendations.

Authors are requested to prepare a revised version of their manuscript as soon as possible. This may ensure fast publication if an article is finally accepted.

Also note that an ORCID ID is mandatory for each author. Kindly provide your ORCID ID in the revised version. If you don't have one, please register by following the steps

Go to <https://orcid.org/register>

Fill in your name and email

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Done, and you will get a 16-digit ORCID ID (like 0000-0002-1825-0097)

Thank you for submitting your work to our journal.

Enter revision:

https://www.editorialsystem.com/juri/article/507649/enter_revision/

Kindest regards,

Dr. Zahid Bhat

Associate Editor

Journal of Undergraduate Research International

Review 1:

Overall, the current form of the manuscript resembles a copy-paste from a thesis or a student project, rather than a structured research article. As such, it is not suitable for publication in its current form. The following major concerns must be addressed prior to resubmission:

Reference inconsistencies: The citation order is incorrect (e.g., jumping from reference [1] to [6], then to [3], and back to [2]). Please carefully revise and reorder the references according to their appearance in the text.

Over-reliance on a single source: Reference [1] is cited more than 12 times. A well-balanced introduction and methodology section should incorporate insights from a broader range of literature. The current approach gives the impression that the manuscript heavily relies on or reproduces the work from Ref [1], even if that is not the case.

Structure of the introduction and conclusion: Avoid using section-like titles within the introduction. Instead, write a coherent narrative that logically presents the background, motivation, and objectives. The same applies to the conclusion, which should be concise, integrative, and reflective of the overall findings.

Formatting issues: Several words throughout the manuscript are unnecessarily written in bold. Please remove these and maintain a consistent, formal formatting style.

Section and subsection numbering: The manuscript contains multiple errors in the section numbering. Please revise to ensure a coherent and consistent structure.

Figures and tables: The numbering of figures and tables is inconsistent, and some titles are missing. For instance, Table 1 and Figure 2 (which should be Figure 1) lack captions. Please revise all figures and tables accordingly.

Feature importance analysis: Although highlighted in the methodology, this analysis is not discussed or presented in the results. Please address this omission.

Prediction results: Figures comparing predicted vs actual values are missing. These are essential for evaluating the performance of your model and should be included.

Writing style: The manuscript should maintain a formal academic tone. Avoid using informal expressions or pronouns such as "we". Revise the text to reflect the appropriate style for a research publication.